

Tutorial No. 8

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Problem Statement : Explain the objectives and principles of software testing and differentiate between verification and validation with suitable examples.

Solution :

Introduction

Software testing is an essential phase of the SDLC. It is the process of evaluating a software system to ensure that it meets specified requirements and functions correctly. Testing helps identify defects, improve quality, and ensure reliability before the software is delivered to users.

Objectives of Software Testing

The main objectives of software testing:

1) Detecting errors or defects.
The primary objective of testing is to find errors, bugs or defects in the software so they can be corrected before the product is released.

2) Ensuring software quality.
Testing ensures that the software meets quality standards in terms of reliability, performance

security, and usability.

- 3.) Verifying Requirements
 - Testing verifies that the developed software meets the functional and non-functional requirements specified by the client
- 4.) Improving Reliability
 - Testing ensures that the system performs correctly under different conditions, to fail without failure
- 5.) Enhancing User Satisfaction
 - A well tested system provides better user experience and increases user confidence in the software product.
- 6.) Preventing future failures.
 - Testing helps detect defects early in the development process, which reduces the cost and risk of fixing them later.

3. Principles of Software Testing

- Software testing follows certain fundamental principles to make the testing effective.
- i.) Testing shows presence of defects
 - Testing can reveal defects in the system, but cannot prove that the software is error free.
- ii.) Exhaustive Testing is Impossible
 - Testing every possible input and scenarios is not feasible, especially for complex systems.
- iii.) Early Testing

Testing activities should begin as early as possible in the SDLC to reduce the cost of fixing defects.

Defect Clustering

A small number of modules in the software usually contain most of the defects.

Pesticide Paradox

If the same set of test cases is repeated multiple times, they may eventually stop finding new defects. Therefore, test cases should be reviewed and updated regularly.

Testing is context dependent.

Testing methods and strategies vary depending on the type of application, such as banking systems, medical systems, or gaming software.

Absence - of - Errors Fallacy

Even if a system has no defects, it may still fail if it does not meet the actual needs of the users.

Verification and Validation

Verification and validation are two important concepts in software testing that ensure the quality of the software product.

Verification: is the process of evaluating the software testing during the development phase to ensure that the product is ~~be~~ being built according to the specified requirements and design documents.

Example:

- Reviewing the design document of a login module to ensure that the product is built according to the username and password validation.

Validation

- It is the process of evaluating the software product to ensure that it meets user's needs and performs the intended ones correctly.

Example:

- Testing the login module by entering valid and invalid usernames and passwords to verify that the system behaves correctly.

5. Difference between Verification and Validation

Basics	Verification	Validation
Definition	Process of checking whether the software is built according to specifications	Process of checking whether software meets user needs
Main Question	Are we building the product right?	Are we building the right product?
Type	Static Process	Dynamic
Activities	Reviews, inspections, walkthroughs	Testing

Performed By	Developers and quality assurance team	Testers and end users.
Stage	Performed during development	Performed after development or during testing phase

Conclusion

Software testing plays a vital role in ensuring the quality and reliability of software systems. The objectives and principles of testing guide the testing process to detect defects efficiently. Verification ensures that the software is developed according to specifications, while validation ensures that the final product meets the needs of users. Together, they help in delivering high-quality software products.

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